Certified AC Robust Secure Sets with GENCO in the Loop: Benders Bounds, Physics-Trajectory Recalibration and Conformal Screening

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***Abstract*—The robust secure set (RSS) gives, for each hour, the range of dispatch that stays secure for every PV and load realisation in a forecast box and in selected contingencies. Its bilevel formulation was solved in the author's dissertation with DC power flow and a big-M KKT MILP. We make the DC solution certified and then make it AC-aware with the GENCO grid foundation model. (i) A two-phase combinatorial Benders decomposition solves all 48 hourly problems of the IEEE 118-bus study to certified optimality and shows that the big-M MILP misreports 20 of them by up to 6.35 MW. (ii) Exact AC power flow shows that 45 of 48 certified DC limits are AC-insecure; a GENCO-in-the-loop loop with exact AC verification and correction cuts removes the violations; DC overstates the secure range by 42.5 % on average and, in hour 18, reports 270.6 MW where the corrected set is empty. (iii) We show that the AC mismatch GENCO measures inside its corrective layers explains 56.3 % of its own error (physics-trajectory recalibration, PTR); wrapped in split-conformal screening it clears 54.4 % of the secure scenarios without exact AC (4.67 % for plain conformal), with a finite-sample bound of 5 % on missed insecure scenarios (observed 0.09 %), also in a topology GENCO never saw. Everything runs from one command with `gridfm-graphkit` and `gridfm-datakit`.**

I. Introduction

A robust secure set (RSS) tells the operator, for every hour, how far the dispatch of controllable units can move while *any* realisation of PV output and load inside a forecast box remains secure in the base case and selected contingencies. Computing it is a bilevel problem: nature chooses the worst PV/load vertex, the operator answers with the best dispatch [1], [2]. Two approximations made it tractable in the first author's dissertation: DC power flow (thermal limits on MW instead of MVA, no losses, no voltages) and a big-M KKT reformulation solved as one MILP.

This entry asks what a grid foundation model adds to such a *certified* workflow. GENCO [3] is a unified neural solver for power flow, OPF and state estimation trained with `gridfm-graphkit` on data from `gridfm-datakit` [4]. We keep certification where it can be had and use GENCO where it is strong: batched, differentiable AC prediction with an internal physics check. Contributions:

1) A two-phase Benders decomposition for the RSS bilevel that needs no big-M on the primal side and returns a certified bound (Theorem 1); on all 48 hourly problems it reproduces the true optimum and exposes reporting errors of the big-M MILP.
2) A GENCO-in-the-loop AC correction: GENCO screens pools of worst-case PV/load vertices in three topologies, exact AC verifies flagged ones, and AC-correction cuts are fed back to the Benders master.
3) *Physics-trajectory recalibration* (PTR): the AC power mismatch that GENCO measures in each of its corrective layers explains a large part of its own error; a small recalibrator on that trajectory, wrapped in split-conformal screening, gives a finite-sample bound on missed insecure scenarios (Theorem 2) while clearing many more secure scenarios than plain conformal screening, also in a topology GENCO never saw.
4) A GENCO-gradient adversary over the PV/load box, evaluated against exact AC power flow.

The complete model (every equation used by the code) and complete proofs of all statements are given in Appendices A–G. Everything is reproducible with one command; no result in this paper is set by hand.

II. Robust Secure Set and Its Certified DC Solution

A. Bilevel model

For hour h, nature chooses the PV and load vector $p = [P_R; P_D]$ in the box $\mathcal{P}_h = [\underline{p}, \overline{p}]$ of (15) and the operator answers with the dispatch $u \geq 0$ of the dispatchable units. The DC line limits in the base topology and in the two contingencies (9), the lossless balance (10) and the ramp-limited unit bounds (12) are stacked as $Ap + Bu \leq b$ (13), and $\mathcal{U}(p) = \{u \geq 0 : Bu \leq b - Ap\}$. With the RSS index $\alpha = d^\top u$ (total output of the non-balancing units), the RSS limits are [1], [2]

$$\overline{\alpha}_h = \min_{p \in \mathcal{P}_h} \max_{u \in \mathcal{U}(p)} d^\top u, \qquad \underline{\alpha}_h = \max_{p \in \mathcal{P}_h} \min_{u \in \mathcal{U}(p)} d^\top u. \quad (1)$$

Theorem 3 (Appendix A) proves that $[\underline{\alpha}_h, \overline{\alpha}_h]$ is *exactly* the set of index values that can be achieved for every realisation in the box, so the set is empty iff $\underline{\alpha}_h > \overline{\alpha}_h$; Lemma 1 proves that

both worst cases sit at vertices of the box. Both limits are linear bilevel programs (19) with an LP follower. The dissertation code solved them as one MILP after replacing the follower by its KKT conditions linearised with $L = 5 \cdot 10^4$ (22); Remark 2 explains why that MILP can report values it does not attain.

B. Two-phase Benders decomposition

Master (phase 1), (24)–(M6): p, u and binaries $z = (z^u, z^\lambda)$ encoding the complementarity pattern,

$$u_k \le \overline{U}_k z_k^u, \qquad b_i - A_i p - B_i u \le S_i(1 - z_i^\lambda), \qquad (2)$$

where $\overline{U}$ and $S_i = b_i - \min_{p,u}(A_i p + B_i u)$ (23) are *exact* bounds from the data: no dual variables and no arbitrary big-M. *Subproblem:* is the dual system $\mathrm{DF}(J_u, J_\ell)$ (25) feasible for the pattern of the master point? If yes, the point is bilevel optimal; if not, a Farkas certificate (Lemma 2) gives the combinatorial cut [7]

$$\textstyle\sum_{k \in C_u}(1 - z_k^u) + \sum_{i \in C_\ell} z_i^\lambda \ge 1. \qquad (3)$$

Robust value-function cut: $d_2^\top u \ge \varphi_R$, where the static LP (27) lets only the balancing unit adapt to p (Lemma 3 proves $\max_{\mathcal{U}(p)} d_2^\top u \ge \varphi_R$ for all p); its dual gives the first incumbent. *Phase 2:* if φ_R does not exist or phase 1 exceeds 15 iterations, the duals are lifted into the master with a bound on the dual side only (29).

Theorem 1 (Certificate; Theorems 4–6)**.** *At every iteration $F(\hat{p}) \le \mathrm{OPT} \le \mathrm{UB}$: the incumbent is the exact follower reaction at its p (always attained) and the master value is a valid bound. Phase 1 terminates finitely; at termination by dual feasibility the solution is optimal. Phase 2 is exact if the dual bound L_d is at least the uniform constant L^* of Lemma 5.*

III. GENCO in the Loop

A. AC-correction loop

For every hour and limit, starting from the certified DC solution: (i) build a pool of PV/load vertices (the Benders worst case, 60 random vertices and GENCO-gradient vertices) and the DC follower response $u(p)$ for each, in all three topologies; (ii) GENCO predicts the security quantities (32) of the whole pool in one batch; (iii) the conformal rule (38) flags scenarios; (iv) flagged scenarios are solved with exact AC power flow (30)–(31); (v) each verified overload derates the DC limit of that branch and topology and AC losses tighten the slack bound (34); Benders is re-solved. Proposition 1 proves that the corrected intervals are nested inside the DC interval, Corollary 1 that an empty corrected set stays empty when the cuts of both limits are merged, and Proposition 2 states what the output certifies.

B. Probing the corrective layers

GENCO's `GNS_heterogeneous` evaluates, inside each of its K corrective layers, the AC mismatch of its intermediate estimate and feeds it back into the bus embeddings. A forward hook on its residual layer records, at no extra cost,

$$\bar{r}^{(l)} = \frac{1}{|\mathcal{N}|}\sum_i \sqrt{(\Delta P_i^{(l)})^2 + (\Delta Q_i^{(l)})^2}, \qquad \rho = \frac{\bar{r}^{(K)}}{\bar{r}^{(1)}}. \quad (4)$$

C. Physics-trajectory recalibration (PTR) and conformal screening

For each security quantity q_k with limit ℓ_k and GENCO prediction $\hat{q}_k$, PTR fits on a *tune* split a regressor of GENCO's error on GENCO-internal features $z = (\hat{q}, \bar{r}^{(1)}, \ldots, \bar{r}^{(K)}, \rho)$ (36) and sets $\tilde{q}_k = \hat{q}_k + g_k(z)$. On an independent *calibration* split [8] of size n the scores $s_k = q_k - \tilde{q}_k$ give the threshold

$$\begin{aligned} c_k &= \text{the } \lceil (n+1)(1-\alpha/3) \rceil\text{-th smallest score,} \\ \text{clear} &\iff \tilde{q}_k + c_k \le \ell_k \quad \forall k. \end{aligned} \qquad (5)$$

Theorem 2 (No-miss guarantee; Theorem 7)**.** *For a test scenario exchangeable with the calibration scenarios and any g_k fitted without them,* $\Pr[\textit{insecure and cleared}] \le \alpha$.

The proof (Appendix E) uses the split-conformal quantile Lemma 6 and a union bound over the three quantities. $g_k \equiv 0$ is plain split-conformal; we also compare a residual-normalised score $(q_k - \hat{q}_k)/(\sigma_k + \beta_k \bar{r}^{(K)})$. PTR uses no hour or topology label, so it applies unchanged to a topology GENCO never saw (where, however, exchangeability is not guaranteed).

D. Gradient adversary

GENCO is differentiable in its inputs. With the dispatch fixed, we ascend the smooth maximum (40) of GENCO-predicted loading over the three topologies by projected sign-gradient steps (41) in $\mathcal{P}_h$ (Lemma 7 bounds the smoothing error by $\tau \log |\mathcal{E}|$), round to the nearest vertex, recompute $u(p)$ and add the point to the pool, where it is verified exactly.

IV. Test System and Implementation

Grid. PGLib-OPF v23.07 IEEE 118-bus case [5] (19 dispatchable units, 35 synchronous condensers, 186 branches with MVA ratings). 20 PV buses grouped in five sites, hourly load and PV profile, PV error $0.3\,h/24$ of capacity and load error $0.03\,h/24$, PV only in hours 7–18, ramp window $\pm 30\,\%$ of capacity around a DC-OPF base dispatch, and the contingencies of the dissertation (one circuit of 69–70 and of 100–103 lost). Generator voltage set-points come from one AC-OPF per hour (pandapower [9]) at the centre of the box, as `gridfm-datakit` does with its OPF step.

Exact AC. `gridfm-datakit` 1.1.0 in PF mode with the `lightsim2grid` Newton–Raphson solver; its Julia OPF step is replaced by our operating-point source (the RSS dispatches). Security = branch apparent power within rating in all three topologies, PQ-bus voltage within $[0.94, 1.06]$ p.u. (0.01 p.u. tolerance; the largest over-voltage in all data is 0.0055 p.u.), slack within capacity. Generator reactive limits are not enforced by the PF and are reported only.

GENCO. The official checkpoints (`gridfm/genco-pf-datakit`) could not be downloaded from our compute environment, so we trained the GENCO architecture (`GNS_heterogeneous`, PowerFlow task, layered physics loss) from scratch with `gridfm-graphkit` 0.9.0.3 on 24,000 datakit scenarios of the RSS operating region (all hours, three topologies, vertex and interior PV/load, vertex and interior dispatches), in a

small CPU configuration ($K = 10$ layers, hidden size 24, 4 heads, 1,077,891 parameters, 15 epochs, about 4 h on two CPU cores). The code loads an official checkpoint unchanged when it is available.

Data for evaluation (all exact AC, never used for training): 2,000 draws of the training distribution; 3,000 draws of the *screening distribution* (DC follower responses at Benders worst-case or random vertices, random line derating $[0.85, 1]$, random topology), split 1,000/1,000/1,000 into tune, calibration and test; and 600 screening draws in a topology GENCO never saw (one circuit of 49–66 lost). All timings: one 2-core CPU, no GPU.

V. Results

Every number below is read from `results/` by `experiments/e7_numbers.py`.

A. Certified DC robust secure set (E1)

All 48 bilevel problems (24 hours, upper and lower limit) were solved to certified optimality (master bound equals the attained value to 10^{-9}). In 47 of them the robust value-function cut is tight and Benders stops after one master solve; one problem (hour 21, upper) needed phase 2. Mean time is 0.513 s against 1.77 s for the big-M MILP of the dissertation code (both with HiGHS on the same CPU), but the more important difference is correctness (Fig. 1b): the MILP's reported objective differs from the value its own leader decision attains in 20 of 48 problems, by up to 6.35 MW (hour 16, lower limit), and in 1 problem its leader decision is not optimal. This is the known failure mode of a fixed big-M in bilevel KKT reformulations [6]; the Benders certificate removes it. The mean DC secure range is 643.7 MW.

B. GENCO accuracy and its corrective-layer trajectory (E3)

The trained GENCO (1,077,891 parameters) predicts the maximum branch loading with mean absolute error 2.75 % (training distribution), 3.14 % (screening distribution) and 2.95 % (unseen topology) of the rating, and voltage magnitudes with mean worst-bus error 0.00448 p.u. The error is *not* uniform (Fig. 2a): GENCO under-predicts a cluster of contingency-100–103 scenarios. Inside GENCO the AC mismatch contracts by more than two orders of magnitude over the $K = 10$ layers, and almost identically in the unseen topology (Fig. 2b). Single scalars of that trajectory are weak error indicators (Spearman 0.225 for $\bar{r}^{(K)}$, AUROC 0.641 for the 10 % largest errors, 0.586 for detecting the unseen topology). The *whole* trajectory is informative (Fig. 2c): a recalibrator that sees only $\bar{r}^{(1)}, \ldots, \bar{r}^{(K)}, \rho$ removes 36.6 % of the error, and PTR (trajectory plus GENCO outputs) removes 56.3 % (MAE 3.11 % $\rightarrow$ 1.36 %) and reduces it in the unseen topology from 2.95 % to 1.84 %.

C. Screening with a guarantee (E3)

Table I and Fig. 3 report the screening over 20 random tune/calibration/test splits of the 3,000 screening scenarios (266 of 1,000 test scenarios are insecure: the screening distribution sits on the RSS boundary by construction). At $\alpha = 0.05$, plain split-conformal screening can clear only 4.67 % of the secure scenarios, because GENCO's error tail forces a wide margin; PTR clears 54.4 % in seen topologies and 55.8 % in the unseen one. The observed rate of insecure-and-cleared scenarios is 0.09 % (worst split 0.2 %), far below the 5 % guarantee.

TABLE I
Secure scenarios cleared without exact AC [%] and observed miss rate [%] (mean of 20 splits; miss = insecure and cleared).

		$\alpha = 0.01$	$\alpha = 0.05$	$\alpha = 0.10$
seen topol.	plain	2.07	4.67	12.8
	normalised	2.94	8.28	16.3
	PTR	**31.9**	**54.4**	**65.1**
	PTR miss	0	0.09	0.305
unseen 49–66	plain	1.57	5.77	10.1
	normalised	4.76	5.32	8.85
	PTR	**31.7**	**55.8**	**70.2**
	PTR miss	0	0.108	0.367

D. AC-corrected robust secure set (E4)

In 45 of 48 hour-limit problems the certified DC limit is *not* AC-secure: some vertex of the PV/load box overloads a branch in AC (worst loading 114.3 % of rating). The GENCO-in-the-loop correction removes all verified violations in at most 3 iterations (29.1 s per limit on average). It moves 28 of 48 limits, by up to 279.0 MW (hour 21, lower limit), and the mean secure range shrinks from 643.7 to 414.2 MW (Fig. 1a, Fig. 4a): DC overstates the range by 42.5 % on average. In hour 18 the corrected lower limit exceeds the upper one, i.e. the corrected model admits no robust secure dispatch although DC reports a range of 270.6 MW. Over the whole run, PTR cleared 34.9 % of the 19,500 pool scenarios (the pools sit on the boundary, so this is lower than in E3); an audit that solved every pool scenario exactly found 1 insecure scenario among the cleared ones (first iteration of one problem; none in any final iteration). The GENCO-gradient adversary found a vertex worse than all Benders and random vertices in 41 of 100 searches, by at most 1.17 % of rating (Fig. 4b); the DC worst-case vertex is usually already close to the AC worst case.

E. Extension: second-life EV batteries (E5)

Placing second-life EV batteries ($P = \mathrm{SoH} \cdot 0.35 \cdot$ PV capacity, 20 sites, 3.6 p.u. in total) as follower resources widens the certified DC range by 561.8 MW on average (361.4–762.8), i.e. 0.777 of the theoretical $2P_{\mathrm{bess}}$. The worst-case points are AC-insecure in 35 of 48 cases with batteries and 39 of 48 without, so the AC correction of E4 is needed with batteries too.

VI. Discussion and Limitations

Where GENCO helps. On a 118-bus case on CPU, the Newton–Raphson solver of `lightsim2grid` needs 0.289 ms per scenario and our GENCO 10.3 ms, so GENCO is

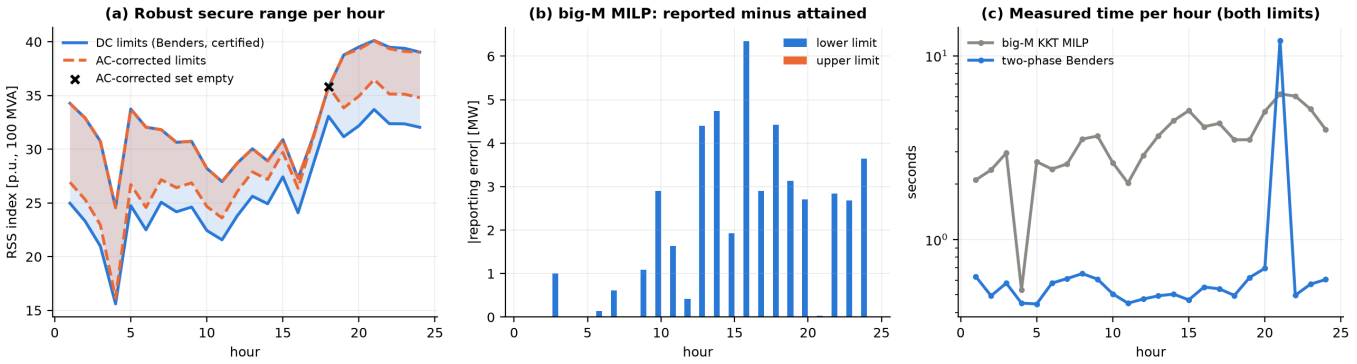


Fig. 1. (a) Certified DC limits per hour and AC-corrected limits from the GENCO-in-the-loop correction (E4). (b) Big-M MILP: reported objective minus value attained by its leader decision. (c) Measured solve time.

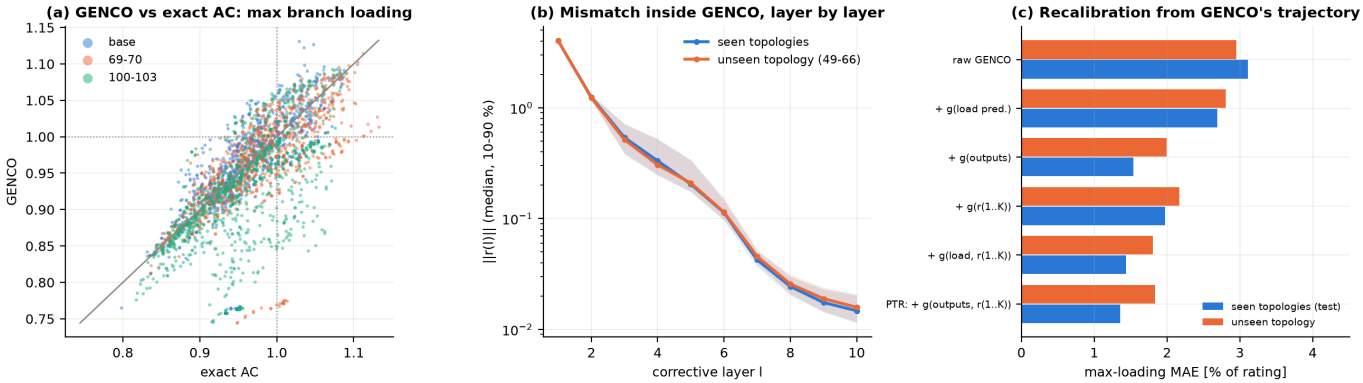


Fig. 2. (a) GENCO vs. exact AC (screening test set), by topology. (b) AC mismatch measured inside GENCO, layer by layer. (c) Ablation: what the recalibrator sees vs. remaining error (fitted on the tune split, evaluated on test and unseen topology).

not a speed-up here; we do not claim one. What GENCO adds is (a) a differentiable surrogate over the uncertainty box, (b) a batched prediction with a self-measured physics trajectory, which PTR turns into the share of exact checks that can be skipped with a guarantee. That share, not raw inference time, is what transfers to settings where the exact check is expensive (larger grids, AC-OPF re-dispatch, reactive limits, many contingencies, GPU batches).

What is certified. The DC bound is certified (Theorem 1). The AC-corrected limit is certified for the corrected DC model, and every pool scenario is AC-secure exactly (flagged) or with probability $1 - \alpha$ (cleared). This is not a certificate over the continuous box in AC; an exact AC robust counterpart remains open. The correction derates active-power limits; all lower-limit problems reached the 0.8 derating floor on some branch, a sign that the MVA overload there is partly reactive, so the corrected limits are conservative (the empty set of hour 18 is a statement about the corrected model, not a proof that no AC-secure dispatch exists).

Honest limitations. The official GENCO checkpoints could not be downloaded from our environment, so a small GENCO was trained from scratch on CPU; a larger model will change all GENCO numbers (the code loads official weights unchanged). Generator reactive limits are not enforced by the power flow. The conformal guarantee needs exchangeability between calibration and use; the unseen-topology results are an empirical check of robustness, not a guarantee. The adversary improved on the Benders and random vertices in 41 of 100 searches, with small gains.

VII. Conclusion

A certified Benders bound, exact AC where it matters, and GENCO with physics-trajectory recalibration where a guarantee allows skipping exact AC together turn the DC robust secure set of the dissertation into an AC-aware one, and expose errors of the big-M MILP on the way. Code, data generation, trained weights, results and an interactive Space are open.

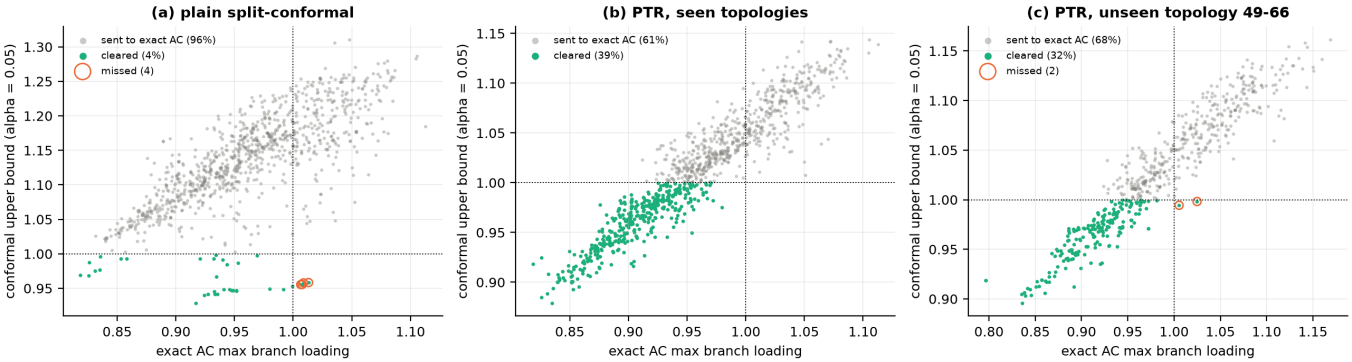


Fig. 3. Conformal upper bound on the maximum branch loading vs. exact AC at $\alpha = 0.05$ (one split). Green: cleared without exact AC. Plain conformal must place almost all scenarios above 1; PTR clears most secure ones, also in the unseen topology.

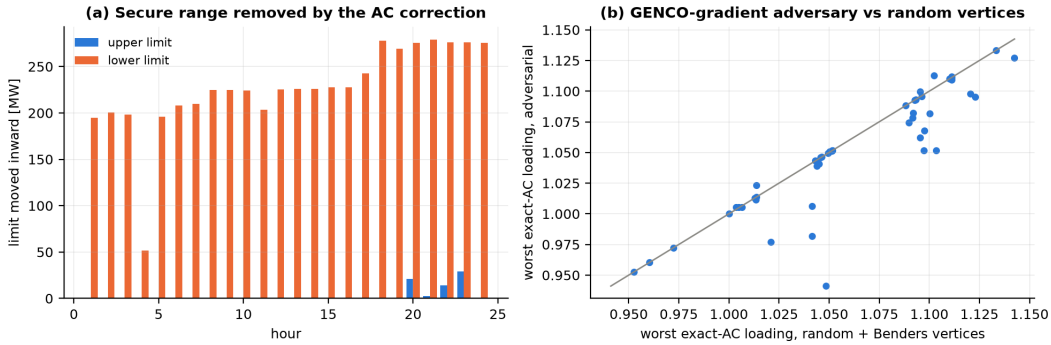


Fig. 4. (a) Secure range removed by the AC correction, per hour and limit (MW the limit moved inward). (b) Worst exact-AC loading at GENCO-adversarial vs. Benders/random vertices (one point per search).

APPENDIX A
DC MODEL OF THE ROBUST SECURE SET

A. Sets, data and decision variables

Buses $\mathcal{N}$ ($|\mathcal{N}| = 118$) with reference bus n_0; branches $\mathcal{L}$ ($|\mathcal{L}| = 186$) with reactance x_ℓ and MVA rating r_ℓ (p.u., base 100 MVA); dispatchable units $\mathcal{G}$ ($|\mathcal{G}| = 19$) with the balancing (slack) unit $g_0 \in \mathcal{G}$ located at n_0; PV buses $\mathcal{R}$ ($|\mathcal{R}| = 20$); load buses $\mathcal{D}$; topologies $s \in \mathcal{S} = \{0, 1, 2\}$ (base case; one circuit of 69–70 lost; one circuit of 100–103 lost). Incidence matrices $C_G \in \{0,1\}^{|\mathcal{N}|\times|\mathcal{G}|}$, C_R, C_D map units, PV injections and loads to buses.

Uncertain vector (nature) $p = [P_R; P_D] \in \mathbb{R}^{n_p}$, $n_p = |\mathcal{R}| + |\mathcal{D}|$. Dispatch (operator) $u \in \mathbb{R}^{n_u}_{\geq 0}$, $n_u = |\mathcal{G}|$ (plus $2|\mathcal{R}|$ battery variables in Appendix G).

B. Power transfer distribution factors

[`grid.ptdf`] Let $C \in \{-1, 0, 1\}^{|\mathcal{L}|\times|\mathcal{N}|}$ be the branch–bus incidence matrix and $X_s = \mathrm{diag}(x^s_\ell)$ with

$$x^s_\ell = \begin{cases} 2x_\ell & \ell \text{ is the corridor of } s \text{ (one of two identical parallel circuits lost)}, \\ x_\ell & \text{otherwise}, \end{cases} \tag{6}$$

since two identical circuits in parallel have half the reactance of one. As in the dissertation data, the corridor rating is kept. With $B^s_f = X_s^{-1}C$, $B^s_{\mathrm{bus}} = C^\top X_s^{-1} C$ and $\mathcal{N}' = \mathcal{N} \setminus \{n_0\}$,

$$H_s[:, \mathcal{N}'] = B^s_f[:, \mathcal{N}'] \big(B^s_{\mathrm{bus}}[\mathcal{N}', \mathcal{N}']\big)^{-1}, \qquad H_s[:, n_0] = 0. \tag{7}$$

The DC branch flows of an operating point in topology s are

$$f_s(p, u) = H_s\big(C_G u + C_R P_R - C_D P_D\big). \tag{8}$$

Because the slack unit sits at n_0, column $H_s C_G[:, g_0] = 0$: the slack unit does not appear in any line constraint (used in Lemma 3).

C. Constraints

[`grid.rss_matrices`] For derating factors $\kappa_s \in (0, 1]^{|\mathcal{L}|}$ ($\kappa \equiv 1$ in the DC study, Appendix D otherwise):

$$-\kappa_s \circ r \leq f_s(p, u) \leq \kappa_s \circ r, \qquad s \in \mathcal{S}, \tag{9}$$

$$\mathbf{1}^\top u + \mathbf{1}^\top P_R - \mathbf{1}^\top P_D = 0, \qquad \text{(lossless balance)} \tag{10}$$

$$\underline{u} \leq u \leq \overline{u}. \tag{11}$$

[`grid.gen_limits`, `rss.dc_opf`] With u^0 the linear-cost DC-OPF dispatch at the box centre and ramp fraction $\rho = 0.3$:

$$\underline{u}_k = \max(P^{\min}_k, u^0_k - \rho P^{\max}_k), \quad \overline{u}_k = \min(P^{\max}_k, u^0_k + \rho P^{\max}_k) \ (k \neq g_0), \quad [\underline{u}_{g_0}, \overline{u}_{g_0}] = [P^{\min}_{g_0}, P^{\max}_{g_0}]. \tag{12}$$

Stacking (9)–(11) row-wise gives the polyhedral system

$$Ap + Bu \leq b, \qquad A = \begin{bmatrix} A_\ell \\ -A_\ell \\ a^\top \\ -a^\top \\ 0 \\ 0 \end{bmatrix}, \ B = \begin{bmatrix} B_\ell \\ -B_\ell \\ \mathbf{1}^\top \\ -\mathbf{1}^\top \\ -I \\ I \end{bmatrix}, \ b = \begin{bmatrix} \kappa \circ r \\ \kappa \circ r \\ 0 \\ 0 \\ -\underline{u} \\ \overline{u} \end{bmatrix}, \tag{13}$$

with $A_\ell = [H_s C_R, \ -H_s C_D]_{s\in\mathcal{S}}$, $B_\ell = [H_s C_G]_{s\in\mathcal{S}}$ (stacked over s), $a = [\mathbf{1}; -\mathbf{1}]$. The follower's feasible set is

$$\mathcal{U}(p) = \{u \in \mathbb{R}^{n_u}_{\geq 0} : Bu \leq b - Ap\}. \tag{14}$$

D. Uncertainty set

[`grid.scenario_box`] For hour h, with PV forecast $\hat{P}_R$, capacity $\bar{P}_R$, forecast load $\hat{P}_D$, $e_{\mathrm{PV}}(h) = 0.3\,h/24$ and $e_D(h) = 0.03\,h/24$:

$$\mathcal{P}_h = \Big\{p : \ \big[\hat{P}_R - e_{\mathrm{PV}}\bar{P}_R\big]_0^{\bar{P}_R} \leq P_R \leq \big[\hat{P}_R + e_{\mathrm{PV}}\bar{P}_R\big]_0^{\bar{P}_R}, \ (1 - e_D)\hat{P}_D \leq P_D \leq (1 + e_D)\hat{P}_D\Big\}, \tag{15}$$

where $[\cdot]_0^{\bar{P}}$ clips to $[0, \bar{P}]$ and $P_R \equiv 0$ for $h \notin [7, 18]$. $\mathcal{P}_h = [\underline{p}, \overline{p}]$ is a box with vertex set $V(\mathcal{P}_h)$.

E. Robust secure set

The RSS index is the total output of the non-balancing units,

$$\alpha = d^\top u, \qquad d_k = 1 \ (k \neq g_0), \ d_{g_0} = 0. \tag{16}$$

Assumption 1. *$\mathcal{U}(p) \neq \emptyset$ for every $p \in \mathcal{P}_h$ (the follower can always serve the load; checked by Benders, which otherwise reports infeasibility).*

Define the follower value functions and the RSS limits

$$\varphi(p) = \max\{d^\top u : u \in \mathcal{U}(p)\}, \qquad \psi(p) = \min\{d^\top u : u \in \mathcal{U}(p)\}, \tag{17}$$

$$\overline{\alpha}_h = \min_{p \in \mathcal{P}_h} \varphi(p), \qquad \underline{\alpha}_h = \max_{p \in \mathcal{P}_h} \psi(p). \tag{18}$$

Theorem 3 (Meaning of the RSS interval)**.** *Under Assumption 1, a value α is achievable for* every *realisation, i.e. $\forall p \in \mathcal{P}_h \ \exists u \in \mathcal{U}(p) : d^\top u = \alpha$, if and only if $\alpha \in [\underline{\alpha}_h, \overline{\alpha}_h]$. In particular the robust secure set is empty if and only if $\underline{\alpha}_h > \overline{\alpha}_h$.*

Proof. Fix p. $\mathcal{U}(p)$ is a non-empty polytope (bounded by (11)), hence compact and convex; its image under the linear map $u \mapsto d^\top u$ is a compact interval, which by (17) equals $[\psi(p), \varphi(p)]$. ($\Leftarrow$) If $\underline{\alpha}_h \leq \alpha \leq \overline{\alpha}_h$, then for every p, $\psi(p) \leq \underline{\alpha}_h \leq \alpha \leq \overline{\alpha}_h \leq \varphi(p)$, so α lies in the image interval and some $u \in \mathcal{U}(p)$ attains it. ($\Rightarrow$) If $\alpha > \overline{\alpha}_h$, take $p^\star \in \arg\min \varphi$ (exists by Lemma 1); then $\alpha > \varphi(p^\star)$ and no $u \in \mathcal{U}(p^\star)$ attains α. The case $\alpha < \underline{\alpha}_h$ is symmetric with $\arg\max \psi$. □

Lemma 1 (Worst cases are vertices)**.** *φ is concave and ψ is convex on $\mathcal{P}_h$. Hence*

$$\overline{\alpha}_h = \min_{p \in V(\mathcal{P}_h)} \varphi(p), \qquad \underline{\alpha}_h = \max_{p \in V(\mathcal{P}_h)} \psi(p).$$

Proof. Let $p_1, p_2 \in \mathcal{P}_h$, $\theta \in [0,1]$, and $u_i \in \mathcal{U}(p_i)$ optimal for $\varphi(p_i)$. Then $u_\theta = \theta u_1 + (1-\theta)u_2 \geq 0$ and $Bu_\theta = \theta B u_1 + (1-\theta) B u_2 \leq b - A(\theta p_1 + (1-\theta)p_2)$, so $u_\theta \in \mathcal{U}(p_\theta)$ and $\varphi(p_\theta) \geq d^\top u_\theta = \theta\varphi(p_1) + (1-\theta)\varphi(p_2)$: φ is concave. The same argument with minimisers shows that ψ is convex. A concave function on a polytope attains its minimum at a vertex (write any point as a convex combination of vertices and use concavity: $\varphi(\sum \theta_j v_j) \geq \sum \theta_j \varphi(v_j) \geq \min_j \varphi(v_j)$); symmetrically a convex function attains its maximum at a vertex. □

APPENDIX B
BILEVEL FORM, KKT CONDITIONS AND THE BIG-M BASELINE

Both limits are instances of the linear bilevel program [`benders.solve`]

$$\mathrm{OPT} = \max_{p \in \mathcal{P}_h, \, u} \ c_1^\top p + d_1^\top u \quad \text{s.t.} \quad u \in \arg\max_{v \in \mathcal{U}(p)} d_2^\top v, \tag{19}$$

with $c_1 = 0$ and $(d_1, d_2) = (-d, d)$ for the upper limit ($\overline{\alpha}_h = -\mathrm{OPT}$) and $(d_1, d_2) = (d, -d)$ for the lower limit ($\underline{\alpha}_h = \mathrm{OPT}$).

Remark 1. Since $d_1 = -d_2$, every follower optimum gives the leader the same value $d_1^\top u = -\max_v d_2^\top v$; optimistic and pessimistic bilevel solutions coincide. For general data we use the optimistic convention: the attained value at p is

$$F(p) = c_1^\top p + \max\{d_1^\top u : \ u \in \mathcal{U}(p), \ d_2^\top u = \max_{v \in \mathcal{U}(p)} d_2^\top v\}, \tag{20}$$

computed by two LPs [`benders.follower_response`].

Follower dual and KKT. The follower LP $\max\{d_2^\top u : Bu \leq b - Ap, \ u \geq 0\}$ has the dual $\min\{\lambda^\top(b - Ap) : \ B^\top\lambda \geq d_2, \ \lambda \geq 0\}$. A feasible u is optimal iff there is λ with

$$\lambda \geq 0, \quad B^\top\lambda \geq d_2, \quad u_k\,(B^\top\lambda - d_2)_k = 0 \ \forall k, \quad \lambda_i\,(b - Ap - Bu)_i = 0 \ \forall i. \tag{21}$$

Let $\mathcal{R} = \{i : B_i \neq 0\}$. Rows with $B_i = 0$ do not enter $B^\top\lambda$; setting $\lambda_i = 0$ there preserves (21), so w.l.o.g. λ is supported on $\mathcal{R}$.

Big-M KKT MILP of the dissertation [`rss.bigm_milp`, `RSS_MILP2.m`]. With $x = (p, u, \lambda)$, $G(x) = \big(0; \ B^\top\lambda - d_2; \ b - Ap - Bu\big)$ and binaries w:

$$\max \ c_1^\top p + d_1^\top u \ \ \text{s.t.} \ \ Ap + Bu \leq b, \ \ 0 \leq x \leq Lw, \ \ 0 \leq G(x) \leq L(\mathbf{1} - w), \ \ p \in \mathcal{P}_h, \tag{22}$$

$L = 5 \cdot 10^4$. If L bounds every component of some optimal KKT pair, (22) is exact.

Remark 2 (Why the big-M MILP misreports). (i) If L is smaller than the components of every optimal KKT pair, (22) cuts the optimum off [6]. (ii) A MILP solver accepts $w_j \leq \varepsilon_I$ as 0 (integrality tolerance, $\varepsilon_I = 10^{-6}$ in HiGHS); then $x_j \leq L\varepsilon_I = 0.05$ while $G_j(x) > 0$ is allowed, i.e. complementarity can be violated by up to $L\varepsilon_I$ per pair. The reported u is then not follower-optimal and the reported objective is not attained. The reporting errors measured in E1 (up to 0.063 p.u.) are of this order. Our method never reports a value that is not attained: the incumbent is always $F(p)$ of (20), computed by LP.

APPENDIX C
TWO-PHASE COMBINATORIAL BENDERS DECOMPOSITION

A. Exact bounds

For $i \in \mathcal{R}$ the largest possible slack over the box is

$$S_i = \max\Big(0,\ b_i - \min_{p\in\mathcal{P}_h,\, 0\le u\le \overline{u}} (A_i p + B_i u)\Big) = \max\Big(0,\ b_i - \textstyle\sum_j \min(A_{ij}\underline{p}_j, A_{ij}\overline{p}_j) - \sum_k \min(0, B_{ik}\overline{u}_k)\Big), \tag{23}$$

and $\overline{U}_k = \overline{u}_k$. Both are computed from the data; no constant is chosen.

B. Phase-1 master

For a finite set $\mathcal{C}$ of cuts, the master problem $\mathrm{MP}(\mathcal{C})$ has variables p, u and binaries $z^u \in \{0,1\}^{n_u}$, $z^\lambda \in \{0,1\}^{|\mathcal{R}|}$:

$$\max\ c_1^\top p + d_1^\top u \tag{24}$$

$$\text{s.t. } Ap + Bu \le b, \quad p \in \mathcal{P}_h, \quad u \ge 0, \tag{M1}$$

$$u_k \le \overline{U}_k\, z^u_k, \qquad k = 1..n_u \tag{M2}$$

$$b_i - A_i p - B_i u \le S_i\,(1 - z^\lambda_i), \qquad i \in \mathcal{R} \tag{M3}$$

$$z^\lambda_a + z^\lambda_j \le 1, \qquad (a,j) \in \Pi \tag{M4}$$

$$d_2^\top u \ge \varphi_R, \qquad \text{if } \varphi_R > -\infty \tag{M5}$$

$$\textstyle\sum_{k\in C_u}(1 - z^u_k) + \sum_{i\in C_\ell} z^\lambda_i \ge 1, \qquad (C_u, C_\ell) \in \mathcal{C} \tag{M6}$$

where Π is the set of opposite row pairs, $[A_a\ B_a] = -[A_j\ B_j]$ with $b_a + b_j > 0$ (e.g. the two sides of a line limit), and φ_R is defined in (27). Interpretation: $z^u_k = 0$ forces $u_k = 0$; $z^\lambda_i = 1$ forces row i to be binding; $z^\lambda_i = 0$ will mean $\lambda_i = 0$.

C. Subproblem, Farkas certificate and cut

For index sets $J_u \subseteq \{1..n_u\}$ (units allowed to be positive) and $J_\ell \subseteq \mathcal{R}$ (rows with positive slack) define the dual system

$$\mathrm{DF}(J_u, J_\ell): \quad \lambda \in \mathbb{R}^{\mathcal{R}}_{\ge 0}, \quad \lambda_i = 0\ (i \in J_\ell), \quad B_{\mathcal{R}}^\top \lambda \ge d_2, \quad (B_{\mathcal{R}}^\top \lambda)_k = d_{2,k}\ (k \in J_u). \tag{25}$$

[`benders._dual_feasible`] Given a master solution $(\hat{p}, \hat{u}, \hat{z})$ with slack $\hat{s} = b - A\hat{p} - B\hat{u}$, the subproblem checks $\mathrm{DF}(J_u(\hat{u}), J_\ell(\hat{s}))$, $J_u(\hat{u}) = \{k : \hat{u}_k > 0\}$, $J_\ell(\hat{s}) = \{i \in \mathcal{R} : \hat{s}_i > 0\}$.

Lemma 2 (Farkas certificate)**.** *Let $\tilde{B} = B_{\mathcal{R}\setminus J_\ell}$ (the rows whose multiplier may be positive). $\mathrm{DF}(J_u, J_\ell)$ is infeasible if and only if there is $y \in \mathbb{R}^{n_u}$ with*

$$\tilde{B}y \le 0, \qquad y_k \ge 0\ (k \notin J_u), \qquad d_2^\top y > 0. \tag{26}$$

Proof. Eliminating $\lambda_{J_\ell} = 0$ and introducing surplus $t \ge 0$ with $t_{J_u} = 0$, DF reads $\{(\lambda, t_{\bar{J}_u}) \ge 0 :\ \tilde{B}^\top\lambda - I_{:,\bar{J}_u} t_{\bar{J}_u} = d_2\}$, $\bar{J}_u$ the complement of J_u. By Farkas' lemma, $\{x \ge 0 : Mx = d_2\}$ is infeasible iff some y has $M^\top y \le 0$ and $d_2^\top y > 0$; with $M = [\tilde{B}^\top,\ -I_{:,\bar{J}_u}]$ this is (26). □

[`benders._farkas_subset`] A certificate also identifies small infeasible subsystems: an elastic LP relaxes each restriction $\lambda_i = 0$ $(i \in J_\ell)$ and $(B^\top\lambda)_k = d_{2,k}$ $(k \in J_u)$ by a non-negative variable and minimises their sum; the restrictions with a positive dual form $C_\ell \subseteq J_\ell$, $C_u \subseteq J_u$. A second LP checks that $\mathrm{DF}(C_u, C_\ell)$ is infeasible; if not, the full sets $C_u = J_u$, $C_\ell = J_\ell$ are used, which is always correct. Smaller sets give stronger cuts.

If $\mathrm{DF}(C_u, C_\ell)$ is infeasible, the *combinatorial Benders cut* (M6) [7] is added. Equivalently $\sum_{k\in C_u} z^u_k - \sum_{i\in C_\ell} z^\lambda_i \le |C_u| - 1$.

D. Robust value-function cut

[`benders.robust_bound`] Let $F \subseteq \{k : d_{2,k} = 0\}$ be a set of balancing columns such that every row of B either has $B_{iF} = 0$, or is an own bound row of one $f \in F$ (with $A_i = 0$), or belongs to the single equality pair (i_1, i_2) with $[A_{i_1}\ B_{i_1}] = -[A_{i_2}\ B_{i_2}]$ (the balance (10)). For the RSS, $F = \{g_0\}$ (the slack unit has a zero PTDF column, Appendix A); if the structure is not met, $F = \emptyset$. Let $[\underline{w}_f, \overline{w}_f]$ be the own-bound interval of $f \in F$, X the other columns, $\ell_F = \sum_f \min(B_{i_1 f}\underline{w}_f, B_{i_1 f}\overline{w}_f)$, $h_F = \sum_f \max(B_{i_1 f}\underline{w}_f, B_{i_1 f}\overline{w}_f)$ and $\overline{A_i p} = \max_{p\in\mathcal{P}_h} A_i p = \sum_j \max(A_{ij}\underline{p}_j, A_{ij}\overline{p}_j)$. Define the *static robust LP*

$$\begin{aligned} \varphi_R = \max_{0\le v\le \overline{u}_X}\ d_{2,X}^\top v \quad \text{s.t.} \quad & B_{iX} v \le b_i - \overline{A_i p} \qquad (i \notin \{i_1, i_2\},\ B_{iF} = 0), \\ & B_{i_1 X} v \le b_{i_1} - \overline{A_{i_1} p} - \ell_F, \qquad B_{i_2 X} v \le b_{i_2} - \overline{A_{i_2} p} + h_F. \end{aligned} \tag{27}$$

Lemma 3 (Robust value-function bound)**.** *For every $p \in \mathcal{P}_h$: $\max\{d_2^\top u : u \in \mathcal{U}(p)\} \ge \varphi_R$.*

Proof. If (27) is infeasible, $\varphi_R = -\infty$ and there is nothing to prove. Let v be optimal in (27) and fix p. Rows without F: $A_i p + B_{iX} v \le \overline{A_i p} + B_{iX} v \le b_i$. Equality pair: write $g = A_{i_1} p + B_{i_1 X} v$, so the pair reads $-b_{i_2} \le g + B_{i_1 F} w \le b_{i_1}$. As w ranges over the box $\prod_f [\underline{w}_f, \overline{w}_f]$, $B_{i_1 F} w$ covers the whole interval $[\ell_F, h_F]$ (a linear image of a box is an interval). The constraints of (27) give $g \le b_{i_1} - \ell_F$ (row i_1) and $-g \le b_{i_2} + h_F$ (row i_2, since $A_{i_2} = -A_{i_1}$, $B_{i_2 X} = -B_{i_1 X}$), i.e. $[-b_{i_2} - g,\ b_{i_1} - g] \cap [\ell_F, h_F] \neq \emptyset$ because $b_{i_1} + b_{i_2} \ge 0$. Pick $w(p)$ in the box with $B_{i_1 F} w(p)$ in this intersection; own-bound rows of F hold by construction. Hence $u = (v, w(p)) \in \mathcal{U}(p)$ and, as $d_{2,F} = 0$, $\max_{u' \in \mathcal{U}(p)} d_2^\top u' \ge d_{2,X}^\top v = \varphi_R$. □

The dual multipliers μ of (27) give the vertex $p_j^{\mathrm{adv}} = \overline{p}_j$ if $(A^\top \mu)_j > 0$, else $\underline{p}_j$, which maximises $\mu^\top A p$ over the box; $F(p^{\mathrm{adv}})$ is the first incumbent (a heuristic; any feasible p is valid).

E. Validity of the master bound

Theorem 4 (Master relaxation)**.** *Let (p, u^*) be bilevel feasible, i.e. $p \in \mathcal{P}_h$ and u^* optimal for the follower at p, and let all cuts in $\mathcal{C}$ be generated as above. Then there is z such that (p, u^*, z) is feasible for* $\mathrm{MP}(\mathcal{C})$*. Consequently* $\mathrm{UB}(\mathcal{C}) := \max \mathrm{MP}(\mathcal{C}) \ge \mathrm{OPT}$.

Proof. Let λ^* satisfy (21) with u^*, supported on $\mathcal{R}$. Set $z_k^u = \mathbb{1}[u_k^* > 0]$ and $z_i^\lambda = \mathbb{1}[\lambda_i^* > 0]$. (M1) holds by feasibility. (M2): if $u_k^* > 0$ then $z_k^u = 1$ and $u_k^* \le \overline{u}_k$ by (11); otherwise $u_k^* = 0$. (M3): if $\lambda_i^* > 0$, complementarity gives zero slack, which is the requirement for $z_i^\lambda = 1$; if $z_i^\lambda = 0$ the slack is at most S_i by (23). (M4): if $\lambda_a^*, \lambda_j^* > 0$ both rows are binding, $[A_a\, B_a](p; u^*) = b_a$ and $-[A_a\, B_a](p; u^*) = b_j$, so $b_a + b_j = 0$, contradicting $(a, j) \in \Pi$. (M5): $d_2^\top u^* = \max_{\mathcal{U}(p)} d_2^\top u \ge \varphi_R$ by Lemma 3. (M6): suppose a cut (C_u, C_ℓ) is violated, i.e. $z_k^u = 1$ for all $k \in C_u$ and $z_i^\lambda = 0$ for all $i \in C_\ell$. Then $u_k^* > 0$ on C_u, so $(B^\top \lambda^*)_k = d_{2,k}$ there by complementarity, and $\lambda_i^* = 0$ on C_ℓ; together with $\lambda^* \ge 0$, $B^\top \lambda^* \ge d_2$ this means λ^* solves $\mathrm{DF}(C_u, C_\ell)$, contradicting how the cut was generated. Hence (p, u^*, z) is master-feasible and its objective, the leader value of (p, u^*), is at most $\mathrm{UB}(\mathcal{C})$. Taking (p, u^*) optimal for (19) gives the claim. □

Lemma 4 (Attained incumbent)**.** *For every $p \in \mathcal{P}_h$, $F(p) \le \mathrm{OPT}$ and $F(p)$ is attained by the bilevel feasible point $(p, u(p))$ returned by* `follower_response`*.*

Proof. $u(p)$ maximises $d_2^\top u$ over $\mathcal{U}(p)$ and, among those maximisers, $d_1^\top u$; so $(p, u(p))$ is feasible for (19) and its value $F(p)$ is at most the optimum. □

Theorem 5 (Certificate and finite termination of phase 1)**.** *Let* $\mathrm{LB}_t = \max_{\tau \le t} F(\hat{p}_\tau)$ *(incumbents, also local search) and* $\mathrm{UB}_t = \min_{\tau \le t} \mathrm{UB}(\mathcal{C}_\tau)$*. Then for every iteration t,*

$$\mathrm{LB}_t \le \mathrm{OPT} \le \mathrm{UB}_t, \tag{28}$$

so $(\mathrm{UB}_t - \mathrm{LB}_t)/\max(1, |\mathrm{UB}_t|)$ is a certified relative gap. In exact arithmetic and without the lift to phase 2, phase 1 terminates after at most $2^{n_u + |\mathcal{R}|}$ iterations, either (a) because $\mathrm{DF}(J_u(\hat{u}), J_\ell(\hat{s}))$ is feasible, in which case $(\hat{p}, \hat{u})$ is optimal and $\mathrm{LB} = \mathrm{UB} = \mathrm{OPT}$, or (b) because the gap is below the tolerance.

Proof. (28) follows from Lemma 4 and Theorem 4 (the cut set only grows, so every $\mathrm{UB}(\mathcal{C}_\tau)$ is a valid bound).

(a) If $\mathrm{DF}(J_u(\hat{u}), J_\ell(\hat{s}))$ has a solution λ, then $\lambda \ge 0$, $B^\top \lambda \ge d_2$, $\lambda_i = 0$ wherever $\hat{s}_i > 0$ and $(B^\top \lambda - d_2)_k = 0$ wherever $\hat{u}_k > 0$: (21) holds, so $\hat{u}$ is follower-optimal at $\hat{p}$ and $(\hat{p}, \hat{u})$ is bilevel feasible with value $\mathrm{UB}(\mathcal{C}_t)$. With (28), $\mathrm{UB}(\mathcal{C}_t) = \mathrm{OPT}$.

Otherwise a cut is added that is violated by the current binary vector $\hat{z}$: consider $J_u^z = \{k : \hat{z}_k^u = 1\} \supseteq J_u(\hat{u})$ (by (M2)) and $J_\ell^z = \{i : \hat{z}_i^\lambda = 0\} \supseteq J_\ell(\hat{s})$ (by (M3)). Restricting a system by fewer equalities/zeros is a relaxation, so $\mathrm{DF}(J_u^z, J_\ell^z)$ feasible would imply $\mathrm{DF}(J_u(\hat{u}), J_\ell(\hat{s}))$ feasible; hence $\mathrm{DF}(J_u^z, J_\ell^z)$ is infeasible. The implementation uses the Farkas subsets of $(J_u(\hat{u}), J_\ell(\hat{s}))$ if $\hat{z}$ violates the resulting cut and otherwise those of (J_u^z, J_ℓ^z); in both cases $C_u \subseteq J_u^z$ and $C_\ell \subseteq J_\ell^z$, so the left-hand side of (M6) is 0 at $\hat{z}$: the cut removes $\hat{z}$. By Theorem 4 it never removes the binary vector of an optimal KKT pair, so the master stays feasible. Each iteration removes at least one of the finitely many binary vectors, which proves finite termination. □

Remark 3. UB is taken from the master solution with relative MIP gap 10^{-8}; the bound is therefore valid up to that tolerance. Tolerances $\hat{u}_k > 10^{-5} \overline{u}_k$ and $\hat{s}_i > 10^{-5} S_i$ define J_u, J_ℓ numerically.

F. Phase 2: dual lifting

If (27) is infeasible or phase 1 exceeds 15 iterations, the master is lifted with dual variables $\lambda \in \mathbb{R}^{\mathcal{R}}_{\ge 0}$ and the constraints

$$B_{\mathcal{R}}^\top \lambda \ge d_2, \qquad B_{\mathcal{R}}^\top \lambda - d_2 \le L_d(\mathbf{1} - z^u), \qquad \lambda \le L_d\, z^\lambda, \tag{29}$$

($L_d = 5 \cdot 10^4$; a big-M on the dual side only), keeping (M1)–(M6).

Lemma 5 (A uniform dual bound exists)**.** *The dual feasible set $\mathcal{D} = \{\lambda \in \mathbb{R}^{\mathcal{R}}_{\geq 0} : B^{\top}_{\mathcal{R}}\lambda \geq d_2\}$ does not depend on p. For every p there is an optimal dual solution at a vertex of $\mathcal{D}$. Hence $L^* = \max_{\lambda \in \mathrm{vert}(\mathcal{D})} \max(\|\lambda\|_\infty, \|B^{\top}\lambda - d_2\|_\infty) < \infty$ bounds an optimal KKT multiplier for every p simultaneously.*

Proof. $\mathcal{D}$ involves only B and d_2. It lies in the non-negative orthant, so it is a pointed polyhedron with finitely many vertices. Under Assumption 1 the follower LP is feasible and bounded (by (11)), so by strong duality the dual $\min_{\lambda \in \mathcal{D}} \lambda^{\top}(b - Ap)$ has a finite optimum, attained at a vertex (fundamental theorem of LP). □

Theorem 6 (Phase-2 exactness)**.** *Every feasible point of the lifted master is bilevel feasible. If $L_d \geq L^*$, the lifted master optimum equals* OPT.

Proof. At a lifted-master point: $z^u_k = 0 \Rightarrow u_k = 0$ by (M2); $z^u_k = 1 \Rightarrow (B^{\top}\lambda - d_2)_k = 0$ by (29); so $u_k(B^{\top}\lambda - d_2)_k = 0$. Likewise $z^{\lambda}_i = 0 \Rightarrow \lambda_i = 0$ and $z^{\lambda}_i = 1 \Rightarrow$ zero slack by (M3). Hence (21) holds and u is follower-optimal: the point is bilevel feasible and its value is $\leq$ OPT. Conversely take an optimal (p, u^*) and a vertex multiplier λ^* (Lemma 5); with z as in the proof of Theorem 4, (M1)–(M6) hold and (29) holds because $\lambda^*_i \leq L^* \leq L_d$ and $(B^{\top}\lambda^* - d_2)_k \leq L^* \leq L_d$ (and $= 0$ when $z^u_k = 1$). So the lifted master value is $\geq$ OPT. □

Phase 2 was needed in 1 of 48 problems (hour 21, upper); its certificate is conditional on $L_d \geq L^*$, whereas phase 1 is unconditional.

G. Algorithm

1) Compute S (23), Π, φ_R and p^{adv} (27); $\mathrm{LB} \leftarrow F(p^{\mathrm{adv}})$, $\mathcal{C} \leftarrow \emptyset$.
2) Solve $\mathrm{MP}(\mathcal{C})$ (lifted if phase 2) $\rightarrow (\hat{p}, \hat{u}, \hat{z})$, $\mathrm{UB} \leftarrow \min(\mathrm{UB}, \mathrm{val})$.
3) $\mathrm{LB} \leftarrow \max(\mathrm{LB}, F(\hat{p}))$; in iteration 1, one-flip local search over vertices.
4) Stop if phase 2, if the gap $\leq 10^{-3}$, or if $\mathrm{DF}(J_u(\hat{u}), J_\ell(\hat{s}))$ is feasible.
5) Otherwise add the Farkas cut (M6) and go to 2; lift (phase 2) after 15 iterations.

APPENDIX D
AC MODEL, SECURITY AND THE AC-CORRECTION LOOP

A. AC power flow

[gridfm-datakit + lightsim2grid] For an operating point (h, s, p, u) the bus voltages $V_i\angle\theta_i$ solve, with the topology-s admittance matrix $Y^s = G^s + jB^s$,

$$P^{\mathrm{inj}}_i = V_i \sum_j V_j \big(G^s_{ij}\cos\theta_{ij} + B^s_{ij}\sin\theta_{ij}\big), \tag{30}$$

$$Q^{\mathrm{inj}}_i = V_i \sum_j V_j \big(G^s_{ij}\sin\theta_{ij} - B^s_{ij}\cos\theta_{ij}\big), \tag{31}$$

where $P^{\mathrm{inj}} = C_G u + C_R P_R - C_D P_D$ at PQ and PV buses (slack injection free), $Q^{\mathrm{inj}} = -C_D(\gamma \circ P_D)$ at PQ buses with the fixed load power factor $\gamma_i = Q^0_{D,i}/P^0_{D,i}$, generator buses at the AC-OPF set-point $V_i = V^{\mathrm{set}}_i(h)$, and $\theta_{n_0} = 0$. The contingency doubles the series impedance $r_\ell + jx_\ell$ and halves the charging susceptance of the corridor (one of two identical circuits lost), consistent with (6); PV injects at unity power factor. Branch $\ell = (i, j)$ in the π-model carries $S^{\mathrm{f}}_\ell = V_i\angle\theta_i\,(I^{\mathrm{f}}_\ell)^*$, $S^{\mathrm{t}}_\ell = V_j\angle\theta_j\,(I^{\mathrm{t}}_\ell)^*$ and $S_\ell = \max(|S^{\mathrm{f}}_\ell|, |S^{\mathrm{t}}_\ell|)$.

B. Security quantities

[`acrss.read_ac`, `genco.evaluate`] With PQ buses $\mathcal{N}_{PQ}$, limits $[\underline{V}_i, \overline{V}_i]$ and tolerance $\tau_V = 0.01$ p.u.,

$$q_{\mathrm{load}} = \max_\ell \frac{S_\ell}{\overline{S}_\ell}, \qquad q_{\mathrm{vlow}} = \max_{i \in \mathcal{N}_{PQ}} (\underline{V}_i - V_i), \qquad q_{\mathrm{vhigh}} = \max_{i \in \mathcal{N}_{PQ}} (V_i - \overline{V}_i), \tag{32}$$

and the scenario is *secure* iff

$$q_{\mathrm{load}} \leq 1, \qquad q_{\mathrm{vlow}} \leq \tau_V, \qquad q_{\mathrm{vhigh}} \leq \tau_V. \tag{33}$$

For verified scenarios the slack output is also checked, $P_{g_0} \leq P^{\max}_{g_0}$.

C. AC-correction update

[`acrss.ac_rss`] At iteration t the DC model uses κ^t and $\overline{u}^t_{g_0}$. For each verified overload ($q_{\mathrm{load},\ell} = S_\ell/\overline{S}_\ell > 1$ in topology s), with $\varepsilon = 0.01$ and floor $\kappa_{\min} = 0.8$,

$$\kappa^{t+1}_{s\ell} = \max\Big(\kappa_{\min},\ \min\Big(\kappa^t_{s\ell},\ \kappa^t_{s\ell}\,\frac{1-\varepsilon}{S_\ell/\overline{S}_\ell}\Big)\Big), \qquad \overline{u}^{t+1}_{g_0} = \overline{u}^t_{g_0} - (1+\varepsilon)\max_{\text{verified}}(P_{g_0} - P^{\max}_{g_0})^+. \tag{34}$$

Proposition 1 (Nesting)**.** *If $\kappa' \le \kappa$ componentwise and $\overline{u}' \le \overline{u}$, then $\mathcal{U}'(p) \subseteq \mathcal{U}(p)$ for all p, $\varphi' \le \varphi$, $\psi' \ge \psi$, and $[\underline{\alpha}'_h, \overline{\alpha}'_h] \subseteq [\underline{\alpha}_h, \overline{\alpha}_h]$. In particular the iterates of (34) produce nested intervals, all inside the DC interval.*

Proof. Only right-hand sides change: the line rows (9) and the bound $\overline{u}_{g_0}$ become tighter, so every $u \in \mathcal{U}'(p)$ satisfies the constraints of $\mathcal{U}(p)$. A maximum over a subset is not larger and a minimum over a subset is not smaller, hence $\varphi' \le \varphi$, $\psi' \ge \psi$, and by (18) $\overline{\alpha}' \le \overline{\alpha}$, $\underline{\alpha}' \ge \underline{\alpha}$. The update (34) is non-increasing in both κ and $\overline{u}_{g_0}$. □

Corollary 1 (Emptiness is robust to merging the cuts)**.** *The two limits are corrected with their own factors $\kappa^{\mathrm{up}}, \kappa^{\mathrm{lo}}$. If $\underline{\alpha}_h(\kappa^{\mathrm{lo}}) > \overline{\alpha}_h(\kappa^{\mathrm{up}})$, then with the merged factors $\kappa = \min(\kappa^{\mathrm{up}}, \kappa^{\mathrm{lo}})$ the robust secure set is also empty: $\underline{\alpha}_h(\kappa) \ge \underline{\alpha}_h(\kappa^{\mathrm{lo}}) > \overline{\alpha}_h(\kappa^{\mathrm{up}}) \ge \overline{\alpha}_h(\kappa)$ by Proposition 1, and Theorem 3 applies.*

Proposition 2 (Termination and what is certified)**.** *Each $\kappa_{s\ell}$ can strictly decrease at most $\lceil \ln \kappa_{\min}/\ln(1-\varepsilon)\rceil = 23$ times before it reaches the floor, and the loop is capped at 6 iterations. If it stops at iteration t because no verified scenario violates, then: (i) α^t is the certified optimum of the corrected DC bilevel (Theorem 5); (ii) every flagged pool scenario is AC-secure, verified by exact power flow; (iii) every cleared pool scenario that is exchangeable with the calibration data is insecure with probability at most α (Theorem 7), and the expected number of insecure cleared scenarios in a pool of N is at most αN; (iv) with the audit enabled (all our experiments) every pool scenario is solved exactly, and a zero audit count means the whole final pool is thermally secure.*

Proof. A strict decrease multiplies $\kappa_{s\ell}$ by $(1-\varepsilon)/q < 1-\varepsilon$ since $q > 1$; after n decreases $\kappa_{s\ell} \le (1-\varepsilon)^n$, which is below $\kappa_{\min}$ for $n \ge \ln \kappa_{\min}/\ln(1-\varepsilon)$. (i) is Theorem 5 applied to the data of iteration t; (ii) is the stopping rule; (iii) is Theorem 7 plus linearity of expectation; (iv) is the definition of the audit. □

Remark 4. (34) is a correction heuristic, not an exact AC robust counterpart: it tightens the *active-power* limit of a branch whose *apparent* power is too high. If the overload is mostly reactive, the floor $\kappa_{\min}$ is reached (it was reached in every lower-limit problem of E4), and the corrected limits are conservative.

APPENDIX E
GENCO TRAJECTORY, RECALIBRATION AND CONFORMAL SCREENING

A. Corrective-layer trajectory

GENCO (`GNS_heterogeneous`) refines a state estimate $\hat{x}^{(l)} = (\hat{V}^{(l)}, \hat{\theta}^{(l)})$ over K layers. In layer l it evaluates the AC mismatch of (30)–(31), $\Delta P_i^{(l)} = P_i^{\mathrm{inj}} - P_i(\hat{x}^{(l)})$, $\Delta Q_i^{(l)} = Q_i^{\mathrm{inj}} - Q_i(\hat{x}^{(l)})$, and feeds it back into the bus embeddings. A forward hook on its residual layer records

$$r_i^{(l)} = \sqrt{(\Delta P_i^{(l)})^2 + (\Delta Q_i^{(l)})^2}, \qquad \bar{r}^{(l)} = \frac{1}{|\mathcal{N}|}\sum_i r_i^{(l)}, \qquad \rho = \bar{r}^{(K)}/\bar{r}^{(1)}. \tag{35}$$

GENCO's predictions $\hat{q}_k$ of (32) use its final state and graphkit's branch-flow operator.

B. Physics-trajectory recalibration (PTR)

Let $z = (\hat{q}_{\mathrm{load}}, \hat{q}_{\mathrm{vlow}}, \hat{q}_{\mathrm{vhigh}}, \hat{q}_{\mathrm{qex}}, \hat{q}_{\mathrm{slack}}, \bar{r}^{(1)}, \ldots, \bar{r}^{(K)}, \rho)$ be GENCO-internal features (no hour, no topology label). On a *tune* set $\mathcal{T}$ solved exactly, fit for each k a gradient-boosted regression ensemble

$$g_k = \arg\min_{g \in \mathcal{F}} \sum_{j \in \mathcal{T}} \big(q_{k,j} - \hat{q}_{k,j} - g(z_j)\big)^2 \quad (300 \text{ depth-3 trees, rate } 0.05,\ \text{subsample } 0.8), \tag{36}$$

and set $\tilde{q}_k = \hat{q}_k + g_k(z)$. Plain conformal uses $g_k \equiv 0$; the residual-normalised variant uses $g_k \equiv 0$ with a scale $\sigma_k + \beta_k \bar{r}^{(K)}$ tuned on $\mathcal{T}$.

C. Split-conformal screening

Let $\ell_{\text{load}} = 1$, $\ell_{\text{vlow}} = \ell_{\text{vhigh}} = \tau_V$, a level $\alpha \in (0,1)$, $a_k = \alpha/3$, and a calibration set $\mathcal{C}$ of n scenarios solved exactly, independent of $\mathcal{T}$. With scale $\sigma_k(z) > 0$ ($\equiv 1$ for plain and PTR),

$$s_{k,j} = \frac{q_{k,j} - \tilde{q}_{k,j}}{\sigma_k(z_j)}, \qquad c_k = \text{the } \lceil (n+1)(1-a_k) \rceil\text{-th smallest of } \{s_{k,j}\}_{j\in\mathcal{C}} \ (+\infty \text{ if } \ > n), \tag{37}$$

and a new scenario is *flagged* (sent to exact AC) iff

$$\exists k: \ \tilde{q}_k + c_k\, \sigma_k(z) > \ell_k; \tag{38}$$

otherwise it is *cleared*.

Lemma 6 (Split-conformal quantile)**.** *Let $S_1, \dots, S_{n+1}$ be exchangeable real random variables, $k = \lceil (n+1)(1-a) \rceil$ and c the k-th smallest of $S_1, \dots, S_n$ ($c = +\infty$ if $k > n$). Then* $\Pr[S_{n+1} > c] \le a$.

Proof. If $k > n$ the event is empty. Otherwise, if $S_{n+1} > c$ then at least k of $S_1, \dots, S_n$ are strictly smaller than S_{n+1}, i.e. $R_{n+1} \ge k$ with $R_i = \#\{j \le n+1 : S_j < S_i\}$. Let $v_{(1)} \le \cdots \le v_{(n+1)}$ be the ordered values. If $R_i \ge k$ then $S_i > v_{(k)}$, which is possible only for indices whose value sits in positions $k+1, \dots, n+1$; hence $\sum_{i=1}^{n+1} \mathbb{1}[R_i \ge k] \le n+1-k$ surely. By exchangeability $\Pr[R_i \ge k]$ is the same for all i, so $(n+1)\Pr[R_{n+1} \ge k] = \mathbb{E}\sum_i \mathbb{1}[R_i \ge k] \le n+1-k$ and $\Pr[S_{n+1} > c] \le \Pr[R_{n+1} \ge k] \le \frac{n+1-k}{n+1} \le a$, because $k \ge (n+1)(1-a)$. □

Theorem 7 (No-miss guarantee)**.** *Let the calibration scenarios and a test scenario be exchangeable (e.g. i.i.d. draws of the screening distribution) and let g_k, σ_k be fixed given $\mathcal{T}$ (fitted without $\mathcal{C}$ and the test scenario). Then*

$$\Pr[\textit{test scenario insecure by (33) and cleared}] \le \alpha. \tag{39}$$

Proof. Condition on $\mathcal{T}$. Then g_k and σ_k are fixed functions, so the scores $s_{k,1}, \dots, s_{k,n}, s_{k,n+1}$ (the last for the test scenario) are exchangeable for each k. If the test scenario is insecure, $q_k > \ell_k$ for some k; if it is cleared, $\tilde{q}_k + c_k\sigma_k \le \ell_k < q_k$, i.e. $s_{k,n+1} > c_k$. Hence

$$\{\text{insecure and cleared}\} \subseteq \bigcup_k \{s_{k,n+1} > c_k\}, \qquad \Pr[\cdot \mid \mathcal{T}] \le \sum_k \Pr[s_{k,n+1} > c_k \mid \mathcal{T}] \le \sum_k a_k = \alpha$$

by the union bound and Lemma 6. Taking the expectation over $\mathcal{T}$ gives (39). □

Corollary 2. *The guarantee holds for* every *recalibrator g_k (including $g_k \equiv 0$ and all ablation variants), because the proof uses only that g_k is fixed before calibration. A better g_k cannot weaken it; it only reduces the flag rate. For N exchangeable scenarios the expected number of insecure-and-cleared ones is at most αN.*

Remark 5. The guarantee is marginal (over the screening distribution), not conditional on the hour or topology. For the unseen topology the exchangeability assumption does not hold; the measured miss rates there are an empirical robustness check.

APPENDIX F
GENCO-GRADIENT ADVERSARY

[`genco.adversarial`] With the dispatch fixed, let $\hat{\ell}_e(p)$, $e \in \mathcal{E}$, be the GENCO-predicted loadings $S_\ell/\overline{S}_\ell$ of all branches in the three topologies (the reactive load moves with P_D at fixed power factor). The objective is the smooth maximum

$$J(p) = \tau \log \sum_{e\in\mathcal{E}} \exp\big(\hat{\ell}_e(p)/\tau\big), \qquad \tau = 0.02. \tag{40}$$

Lemma 7 (Smooth maximum)**.** $\max_e \hat{\ell}_e \le J \le \max_e \hat{\ell}_e + \tau \log|\mathcal{E}|$, *and* $\nabla_p J = \sum_e \pi_e \nabla_p \hat{\ell}_e$ *with* $\pi = \mathrm{softmax}(\hat{\ell}/\tau)$ *in the simplex.*

Proof. With $m = \max_e \hat{\ell}_e$: $e^{m/\tau} \le \sum_e e^{\hat{\ell}_e/\tau} \le |\mathcal{E}| e^{m/\tau}$; take $\tau \log$. The gradient follows from the chain rule, $\partial J/\partial \hat{\ell}_e = e^{\hat{\ell}_e/\tau}/\sum_{e'} e^{\hat{\ell}_{e'}/\tau} = \pi_e$. □

The update is projected sign-gradient ascent on the box,

$$p \leftarrow \Pi_{\mathcal{P}_h}\big(p + \eta\,(\overline{p} - \underline{p}) \circ \operatorname{sign} \nabla_p J(p)\big), \qquad \Pi_{\mathcal{P}_h}(y) = \min(\max(y, \underline{p}), \overline{p}), \ \eta = 0.25, \tag{41}$$

for 6 steps from the riskiest pool points; the result is rounded to the nearest vertex, $p_j \leftarrow \overline{p}_j$ if $p_j > \frac{1}{2}(\underline{p}_j + \overline{p}_j)$ else $\underline{p}_j$ (by Lemma 1 the DC worst case is a vertex), the follower response $u(p)$ is recomputed and the point is verified by exact AC. $\Pi_{\mathcal{P}_h}$ is the Euclidean projection onto a box (separable). The adversary is a heuristic; nothing is certified from it without exact verification.

APPENDIX G
SECOND-LIFE EV BATTERIES

[`bess.bess_power`, `rss._add_bess`] At each PV bus $j \in \mathcal{R}$ a battery of power $P_j^{\mathrm{b}} = \mathrm{SoH}_j \cdot 0.35 \cdot \bar{P}_{R,j}$ adds discharge and charge variables $0 \le u_j^{\mathrm{dis}}, u_j^{\mathrm{ch}} \le P_j^{\mathrm{b}}$ to the follower. Its net injection $u^{\mathrm{dis}} - u^{\mathrm{ch}}$ enters (8) through C_R and the balance (10); it does not enter the index, $d_{\mathrm{dis}} = d_{\mathrm{ch}} = 0$.

Proposition 3 (Batteries can only widen the RSS)**.** $\overline{\alpha}_h^{\mathrm{b}} \ge \overline{\alpha}_h$ *and* $\underline{\alpha}_h^{\mathrm{b}} \le \underline{\alpha}_h$.

Proof. For every $u \in \mathcal{U}(p)$, $(u, 0, 0)$ is feasible with batteries (zero battery variables leave all rows unchanged), with the same index. Hence the index range at each p can only grow, $\varphi^{\mathrm{b}} \ge \varphi$ and $\psi^{\mathrm{b}} \le \psi$, and (18) gives the claim. □

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